

● HARD

● CRAFT AND STRUCTURE

Craft and Structure

10 questions · ~15 min

06

Q1

CRAFT AND STRUCTURE

Text 1

What factors influence the abundance of species in a given ecological community? Some theorists have argued that historical diversity is a major driver of how diverse an ecological community eventually becomes: differences in community diversity across otherwise similar habitats, in this view, are strongly affected by the number of species living in those habitats at earlier times.

Text 2

In 2010, a group of researchers including biologist Carla Cáceres created artificial pools in a New York forest. They stocked some pools with a diverse mix of zooplankton species and others with a single zooplankton species and allowed the pool communities to develop naturally thereafter. Over the course of four years, Cáceres and colleagues periodically measured the species diversity of the pools, finding—contrary to their expectations—that by the end of the study there was little to no difference in the pools’ species diversity.

Based on the texts, how would Cáceres and colleagues (Text 2) most likely describe the view of the theorists presented in Text 1?

- A. It is largely correct, but it requires a minor refinement in light of the research team’s results.
- B. It is not compelling as a theory regardless of any experimental data collected by the research team.
- C. It may seem plausible, but it is not supported by the research team’s findings.
- D. It probably holds true only in conditions like those in the research team’s study.

“How lifelike are they?” Many computer animators prioritize this question as they strive to create ever more realistic environments and lighting. Generally, while characters in computer-animated films appear highly exaggerated, environments and lighting are carefully engineered to mimic reality. But some animators, such as Pixar’s Sanjay Patel, are focused on a different question. Rather than asking first whether the environments and lighting they’re creating are convincingly lifelike, Patel and others are asking whether these elements reflect their films’ unique stories.

Which choice best describes the function of the underlined question in the text as a whole?

- A. It reflects a primary goal that many computer animators have for certain components of the animations they produce.
- B. It represents a concern of computer animators who are more interested in creating unique backgrounds and lighting effects than realistic ones.
- C. It conveys the uncertainty among many computer animators about how to create realistic animations using current technology.
- D. It illustrates a reaction that audiences typically have to the appearance of characters created by computer animators.

Some scientists have suggested that mammals in the Mesozoic era were not a very _____ blank group, but paleontologist Zhe-Xi Luo's research suggests that early mammals living in the shadow of dinosaurs weren't all ground-dwelling insectivores. Fossils of various plant-eating mammals have been found in China, including species like *Vilevolodon diplomylos*, which Luo says could glide like a flying squirrel.

Which choice completes the text with the most logical and precise word or phrase?

- A. predatory
- B. obscure
- C. diverse
- D. localized

Text 1

Dominique Potvin and colleagues captured five Australian magpies (*Gymnorhina tibicen*) to test a new design for attaching tracking devices to birds. As the researchers fitted each magpie with a tracker attached by a small harness, they noticed some magpies without trackers pecking at another magpie's tracker until it broke off. The researchers suggest that this behavior could be evidence of magpies attempting to help another magpie without benefiting themselves.

Text 2

It can be tempting to think that animals are deliberately providing help when we see them removing trackers and other equipment from one another, especially when a species is known to exhibit other cooperative behaviors. At the same time, it can be difficult to exclude the possibility that individuals are simply interested in the equipment because of its novelty, curiously pawing or pecking at it until it detaches.

Based on the texts, how would the author of Text 2 most likely respond to the researchers' perspective in Text 1 on the behavior of the magpies without trackers?

- A. That behavior might have been due to the novelty of the magpies' captive setting rather than to the novelty of the tracker.
- B. That behavior likely indicates that the magpies were deliberately attempting to benefit themselves by obtaining the tracker.
- C. That behavior may not be evidence of selflessness in *Gymnorhina tibicen* because not all the captured magpies demonstrated it.
- D. That behavior might be adequately explained without suggesting that the magpies were attempting to assist the other magpie.

Historians have argued that a crucial component of the Civil Rights Movement's success in the 1960s was the Southern Christian Leadership Conference's Citizen Education Program (CEP), which invited promising activists from across the South to its one-week training sessions in Dorchester, Georgia. Led by experienced organizers such as Dorothy Cotton and Septima Clark, CEP attendees—more than 7,000 in all—participated in workshops on topics ranging from public speaking to legal doctrine before returning home and using their newly acquired knowledge to spearhead local civil rights initiatives.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It underscores the extent of the CEP's impact on the Civil Rights Movement of the 1960s.
- B. It illustrates the CEP organizers' efforts to educate participants on a wide variety of topics.
- C. It suggests that CEP attendees held a diverse array of opinions about the Southern Christian Leadership Conference's political philosophy.
- D. It establishes that criticism of the CEP was limited to a few individuals in the Southern Christian Leadership Conference.

Within baleen whale species, some individuals develop an accessory spleen—a seemingly functionless formation of splenic tissue outside the normal spleen. Given the formation’s greater prevalence among whales known to make deeper dives, some researchers hypothesize that its role isn’t _____ blank; rather, the accessory spleen may actively support diving mechanisms.

Which choice completes the text with the most logical and precise word or phrase?

- A. replicable
- B. predetermined
- C. operative
- D. latent

Text 1

Chemically analyzing bulk rock-core samples from Australia's Mount McRae Shale, Ariel Anbar et al. found a transient but significant increase in molybdenum (Mo) at a point corresponding to roughly 2.5 billion years ago (Ga). On Earth, Mo is released mainly through oxidative weathering of minerals; Anbar et al. therefore concluded that atmospheric oxygen briefly increased 2.5 Ga, then returned to its earlier negligible level.

Text 2

Sarah Slotznick et al. reexamined the Mount McRae Shale core. Since chemically analyzing bulk samples can occlude contextual details, Slotznick et al. also employed high-resolution microscopy, which revealed volcanic debris—a known Mo host—around 2.5 Ga and microfractures in the surrounding matrix. The researchers assert that fluid could have reached the debris through the microfractures and initiated oxidative weathering long after debris deposition.

Based on the texts, Anbar et al. (Text 1) and Slotznick et al. (Text 2) would most likely disagree about the answer to which question about the portion of the Mount McRae Shale rock-core corresponding to 2.5 Ga?

- A. Does the increase in Mo in that portion suggest that atmospheric oxygen levels were not negligible before 2.5 Ga?
- B. Is the increase in Mo in that portion indicative of an increase in atmospheric oxygen dating to the same time?
- C. Is the increase in Mo in that portion attributable to oxidative weathering of the material found in that portion?

- D. Did chemically analyzing bulk samples lead to a false impression that there is an increase in Mo in that portion?

In response to concerns that some recent financial crises were exacerbated by consumers misunderstanding risks associated with credit cards, loans, and other financial products, policymakers in many countries have instituted risk-disclosure requirements on sellers of those products. Enrique Seira et al. investigated a variety of risk-disclosure messages sent to thousands of credit card customers and found that the messages had only small and short-lived effects on behavior. Seira et al. asserted that such effects may nevertheless be worth pursuing, given the negligible cost of messaging.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It notes a factor that led Seira et al. to not dismiss risk-disclosure messaging altogether despite their evidence of its limited utility.
- B. It acknowledges a type of risk-disclosure messaging that Seira et al. may not have fully accounted for in their study.
- C. It describes a consideration that explains why Seira et al. recommended risk-disclosure messaging even though its effects may be small relative to its costs.
- D. It points out a circumstance that Seira et al. conceded may make risk-disclosure messaging more effective than their study suggests.

One popular theory of the origin of the Moon, the “big whack,” posits that a protoplanet called Theia collided with Earth, flinging debris into orbit that eventually coalesced into the Moon. Until recently, Theia was _____ blank, but researcher Qian Yuan and colleagues now claim to have identified pieces of the protoplanet in the lowermost section of Earth’s mantle.

Which choice completes the text with the most logical and precise word or phrase?

- A. desultory
- B. spurious
- C. veritable
- D. notional

Text 1

When the 50-second-long film *Arrival of the Train*—which depicts what its title says, a train pulling into a station—was first shown publicly in 1896, spectators, naïve to the new medium of film and seeing a train appearing to come directly at them, leaped from their seats and fled the room. This moment marks a major cultural shift: a new way of representing and seeing the world had arrived with that train, and nothing would ever be the same.

Text 2

The fact that there is no contemporary evidence that the first audience of *Arrival of the Train* was alarmed has not stopped the story from becoming canonical, even among film historians. But that phenomenon itself is highly revealing. Our belief that the coming of film was transformative is so strong that we invented and keep retelling a founding myth that divides cultural history into a (naïve) “before” and (sophisticated) “after.”

Based on the texts, the author of Text 2 would most likely agree with which statement about the description of the first showing of *Arrival of the Train* in Text 1?

- A. It is more motivated by a perception of the significance of the invention of film than by facts.
- B. It expresses a view about the transformative nature of film that film historians tend to regard as naïve.
- C. It is not based on evidence and reflects film historians’ belief that they are more sophisticated than today’s audiences.
- D. It reflects a misconception that is widely believed because it captures early film audiences’ sense that the new medium was alarming.